SCIIENet: Shared and Complementary Information Interaction Enhancement Network for Self-Supervised Multimodal Remote Sensing Image Classification

Pengcheng Guo, Ying Liang, Junhuan Peng, Yuebin Wang, *Member, IEEE*, Zhen Wang, Huiwei Su, Guodong Liu, and Yasong Mi

***Abstract*— Employing multimodal remote sensing images (MRSIs) enhances ground object identification, yet the heterogeneity of MRSIs can increase the risk of model overfitting, particularly when training samples are limited. Self-supervised methods, such as masked autoencoder (MAE) reconstruction or contrastive learning (CL), efficiently extract features from MRSIs with minimal reliance on labeled samples. Current methods, however, have not adequately considered both the shared and complementary features of MRSIs during the pretraining process, limiting feature representation and classification performance. In order to address these limitations, we propose a self-supervised pretraining framework called shared and complementary information interaction enhancement network (SCIIENet) for hyperspectral images (HSIs) and light detection and ranging (LiDAR)/synthetic aperture radar (SAR) data joint classification. Specifically, the framework incorporates a decoupling-interaction-enhancement approach through a triple-branch MAE architecture. The triple-branch architecture separately captures the spectral features of HSIs that provide complementary information relative to LiDAR/SAR data, as well as the spatial features of HSIs and LiDAR/SAR data. We also introduce an interaction enhancement strategy (IES) that uses hierarchical CL (HCL) to bridge significant gaps in MRSI to emphasize the shared spatial information of MRSIs and incorporates a two-stage cross-modal fusion encoder (TCFE) to integrate the shared and complementary spatial information of different modalities. We, moreover, propose multistage skip connections (MSCs) for the reconstruction, which effectively preserve complementary spatial information. Extensive experiments on four datasets show that our method significantly improves classification accuracy compared to several state-of-the-art methods.**

Received 17 October 2024; revised 29 January 2025 and 26 March 2025; accepted 25 April 2025. Date of publication 2 May 2025; date of current version 15 May 2025. This work was supported by the Ministry of Natural Resources Key Laboratory of Natural Resources Investigation, Monitoring and Protection Open Fund Project, under Grant 2023-B01. *(Corresponding authors: Zhen Wang; Huiwei Su.)*

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Digital Object Identifier 10.1109/TGRS.2025.3566405

***Index Terms*— Interaction enhancement, multimodal remote sensing image (MRSI) classification, self-supervised learning, shared and complementary information.**

I. Introduction

IN RECENT years, with the development of remote sensing technology, a wealth of multimodal remote sensing images (MRSIs) has been acquired, including multispectral imagery (MSI), hyperspectral imagery, light detection and ranging (LiDAR) imagery, and synthetic aperture radar (SAR) imagery. Among them, hyperspectral images (HSIs) [1], [2], [3] provide detailed spectral information that is valuable for distinguishing various ground objects. Relying solely on the monomodal data, however, has limitations. Objects with similar spectral characteristics, such as roofs, facades, and roads, can be more effectively distinguished using LiDAR [4], [5], [6] or SAR data [7], [8]. MRSIs can, therefore, leverage their specific information to complement other data [9] and play a crucial role in land cover classification [10], environmental monitoring [11], and urban planning [12].

Deep-learning (DL) methods [13], [14], [15], and [16] have demonstrated superior capabilities in MRSI processing, where feature fusion is a key step. Stacking or aggregating features is a straightforward fusion method that has been widely adopted [17], [18]. This approach, however, often fails to facilitate comprehensive interactions among features due to inadequate consideration of the heterogeneity among MRSIs. In order to further augment feature interactions, various fusion strategies, including multiscale feature fusion [19], [20], weight-sharing [21], [22], and attention-based mechanisms [23], [24], have been developed. These fusion results demonstrate that fully using the shared and complementary features by information interaction between different modalities is important for MRSI classification.

For MRSI classification tasks [25], [26], [27], [28], due to the heterogeneity among multimodal data, models are more susceptible to overfitting when the labeled data is insufficient. Recent advancements in computer vision have demonstrated that self-supervised learning [29], [30] can effectively extract generic feature representations from unlabeled data during the

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pretraining stage. These extracted features can subsequently be transferred to downstream classification tasks, enhancing performance through fine-tuning. Existing self-supervised methods are primarily categorized into contrastive-based and reconstruction-based strategies.

Several contrastive-based studies have been employed for MRSI classification [31], [32]. During the pretraining stage, contrastive learning (CL) enables interaction between different modalities by aligning heterogeneous features. It allows the two modalities to mutually enhance each other through shared features, particularly when data from one modality is poorly captured and crucial information is missing. Several studies [33], [34] have, however, found that CL is insufficient for extracting complementary features and have proposed cross-modal fusion modules to integrate such features during the fine-tuning stage; nevertheless, the extraction of these complementary features may not be fully optimized during pretraining, which limits classification performance. CL may, moreover, risk omitting complementary information if aligned features become excessively similar. Such omissions can constrain the performance of downstream tasks.

Reconstruction using the masked autoencoder (MAE) [35] based on generative strategies has also been applied to MRSI classification [36], [37], [38]. During the pretraining stage, this approach used a transformer encoder–decoder architecture to learn image representations by masking the original image and forcing the model to reconstruct it. Reconstruction facilitates effective information interaction, enabling the fusion of features more compactly while preserving the specific features of different modalities, thereby benefiting downstream tasks. Reconstruction, however, primarily focuses on integrating the features of all modalities, neglecting to deeply explore the complex relationships and feature interactions across them. Such limitations could restrict the performance of feature representation.

In summary, shared and specific or complementary features play crucial roles in MRSI classification. Existing self-supervised studies, however, have not fully addressed both of them during the pretraining stage. In order to address this limitation, we propose the shared and complementary information interaction enhancement network (SCIIENet) as a self-supervised pretraining framework for MRSI classification. The proposed SCIIENet framework incorporates a decoupling-interaction-enhancement approach through a triple-branch architecture consisting of spectral (Spe), spatial (Spa), and LiDAR/SAR (L) branches. The Spe branch independently processes complementary spectral information, while the masked reconstruction is applied across all branches to ensure maximal information preservation; furthermore, we introduce an interaction enhancement strategy (IES) that incorporates hierarchical CL (HCL) and a two-stage cross-modal fusion encoder (TCFE) for the Spa and L branches, which effectively enhances the interaction between shared and complementary spatial information. Additionally, multistage skip connections (MSCs) are implemented across all branches to maintain complementary spatial information. We evaluate the SCIIENet on four MRSI datasets. The experimental results demonstrate that our method outperforms several state-of-the-art methods.

The main contributions of this work can be summarized as follows.

1) We propose a fine-grained decoupling of MRSIs into shared spatial information, complementary spatial information, and complementary spectral information. This refined decomposition enables more precise modeling of these information types during the pretraining phase, resulting in enhanced representation of multimodal data.
2) We introduce an IES in SCIIENet that fully considers the heterogeneity of MRSIs. This strategy employs HCL to improve the framework's ability to narrow the large gap of MRSI by aligning multiple intermediate representations. Additionally, compared to other fusion methods that focus solely on complementary information, the TCFE more effectively integrates both complementarity and shared information through an initial fusion step, followed by a subsequent feature separation step.
3) We design MSCs for each branch of SCIIENet to enhance the reconstruction performance by leveraging multiple intermediate features, thereby better preserving all the information. This design, when combined with HCL, enables SCIIENet to operate in a dynamic and adaptive manner, progressively extracting shared features while preserving complementary information throughout the pretraining.

The remainder of this article is organized as follows. In Section II, we review related supervised methods for MRSIs, as well as self-supervised methods for HSIs and MRSIs. A detailed description of SCIIENet is provided in Section III. Experiments are presented in Section IV. The conclusions are drawn in Section V.

II. Related Works

A. Supervised Learning in MRSI Classification

Supervised learning through DL has attracted extensive research attention, leading to the development of various multimodal data fusion models for MRSI classification. Gao et al. [39] proposed an adaptive feature fusion network that integrated three feature-level fusion strategies, such as concatenation, maximization, and summation, to combine features. Simple fusion methods, however, often fail to fully exploit the interaction information, which limits classification performance. Yao et al. [40] processed MRSI patches using parallel branches of position-shared vision transformer (ViT) and employed a cross-modality attention module for fusion. Additionally, Song et al. [41] proposed a deformable convolution-based height information-guided fusion module and a modality separation refinement module to sequentially extract modality-shared and modality-specific features. By employing the designed fusion approaches, classification performance can be enhanced.

In order to enhance the fusion of spatial and spectral information in HSIs, some studies have employed a triple-branch architecture that separately extracts spectral, spatial, and LiDAR/SAR features. Wang et al. [42] proposed a triple-branch convolutional neural network (CNN) backbone,

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while Xue et al. [43] developed a triple-branch ViT architecture for MRSI classification. These studies demonstrated that a multibranch structure effectively facilitates improved feature fusion.

B. Self-Supervised Learning in HSI Classification

Recently, self-supervised approaches have gained attention in remote sensing image classification, primarily focusing on HSI classification, while only a limited number of studies have addressed MRSI classification; therefore, self-supervised methods developed for HSI classification are also introduced here for reference.

For contrastive-based methods, Hou et al. [44] used random Gaussian noise for data augmentation to construct positive and negative sample pairs and applied CL to these pairs for pretraining. Recognizing that classification performance is influenced by augmentation techniques, Ayuba et al. [45] proposed a specific augmentation method for HSIs to generate contrastive samples. Qin et al. [46] implemented CL by spatial neighborhood-based data augmentation. Focusing solely on spatial diversity while neglecting spectral diversity in HSIs can, however, lead to incomplete feature exploration and poor generalization to downstream tasks. In order to address this issue, Xia et al. [47] proposed a two-stream CL network to separately capture spatial and spectral features for pretraining and then employed a multilevel fusion network to integrate these features during the fine-tuning stage.

In reconstruction-based methods, Liu et al. [48] introduced a self-supervised method involving masked spectral reconstruction. Ibañez et al. [49] developed a masked auto-encoding spectral–spatial transformer, which learned feature representations through grouped mask reconstruction. In order to further enhance performance, Mohamed et al. [50] employed separate masked spectral and spatial reconstruction for pretraining.

Because the two self-supervised methods extract features based on different principles, some studies have explored their combination to achieve more robust feature representations. Huang et al. [51] and Cao et al. [52] proposed similar architectures, using reconstruction for HSIs and CL to encode feature pairs from both masked and original images during the pretraining stage. Their studies demonstrated that integrating the two approaches results in improved performance.

C. Self-Supervised Learning in MRSI Classification

In contrast to HSI classification, both contrastive-based and reconstruction-based self-supervised methods have been employed for MRSI classification. To the best of our knowledge, no method, however, has yet integrated these two approaches.

For contrastive-based methods, Xue et al. [53] constructed multiple views from MRSIs and learned high-level feature representations from each view through CL. The method, however, primarily focused on intermodal relationships while neglecting the cross-modal effects. In order to leverage shared information, Wang et al. [33] applied CL to feature pairs from HSIs and LiDAR data, whereas, Feng et al. [34] applied CL separately for each modality as well as in a cross-modal manner. In order to prevent the loss of complementary information, both methods integrated fusion modules during the fine-tuning phase. They overlooked that the interaction of complementary information across different modalities during the pretraining phase can, however, further enhance performance.

In reconstruction-based methods, Xue et al. [36] constructed multiple observed views from MRSIs and stacked them for masked spatial reconstruction. In order to better investigate the spectral features, Lin et al. [37] stacked MRSIs and applied masked spatial and spectral reconstruction separately. In order to exploit the shared information of the interactions between modalities, Chan-To-Hing and Veeravalli [38] introduced the Fus-MAE, which used cross-attention to perform early feature-level fusion for multimodal data before reconstruction. Because of the heterogeneity of MRSIs, the differences in feature spaces may, nevertheless, limit early data fusion performance.

Given these observations, we would leverage both shared and complementary information by integrating contrastive and reconstruction-based self-supervised learning for MRSI classification. The CL aligns features to minimize the heterogeneity of MRSI by shared information, whereas reconstruction preserves complementary information. Additionally, an IES is employed to better address the heterogeneity of MRSIs and fuse shared and complementary information. Meanwhile, MSCs are used to further improve reconstruction performance and preserve complementary information.

III. Methodology

A. Overview

The self-supervised pretraining framework, SCIIENet, is illustrated in Fig. 1. SCIIENet contains three MAE branches: Spe, Spa, and L, which are used to extract features from each modality and reconstruct the corresponding modality. The IES, incorporating HCL and TCFE, is employed in Spa and L branches. In order to better maintain feature completeness, MSCs based on an attention mechanism are employed for each branch. After pretraining, all the encoders of SCIIENet are employed for fine-tuning, as depicted by the solid-line stereoboxes in Fig. 1.

B. Triple-Branch Self-Supervised Pretraining Framework

Considering the importance of using both shared and complementary information for MRSI classification, we aim to develop a self-supervised pretraining framework that preserves specific complementary information while closely aligning shared information. In the framework, the Spe branch is proposed for spectral reconstruction to avoid the loss of complementary spectral features. CL is used to enable the network to focus on shared spatial information, while the reconstruction is used to preserve complementary spatial information. The details of the framework are presented as follows.

The original input of the framework is a pair of HSI patch $\mathbf{X} \in \mathbb{R}^{S\times S\times B_1}$ and LiDAR/SAR patch $\mathbf{X}_{\mathrm{L}} \in \mathbb{R}^{S\times S\times B_2}$ covering the same region. S is the sliding window size. B_1 is the band number after principal component analysis (PCA) and

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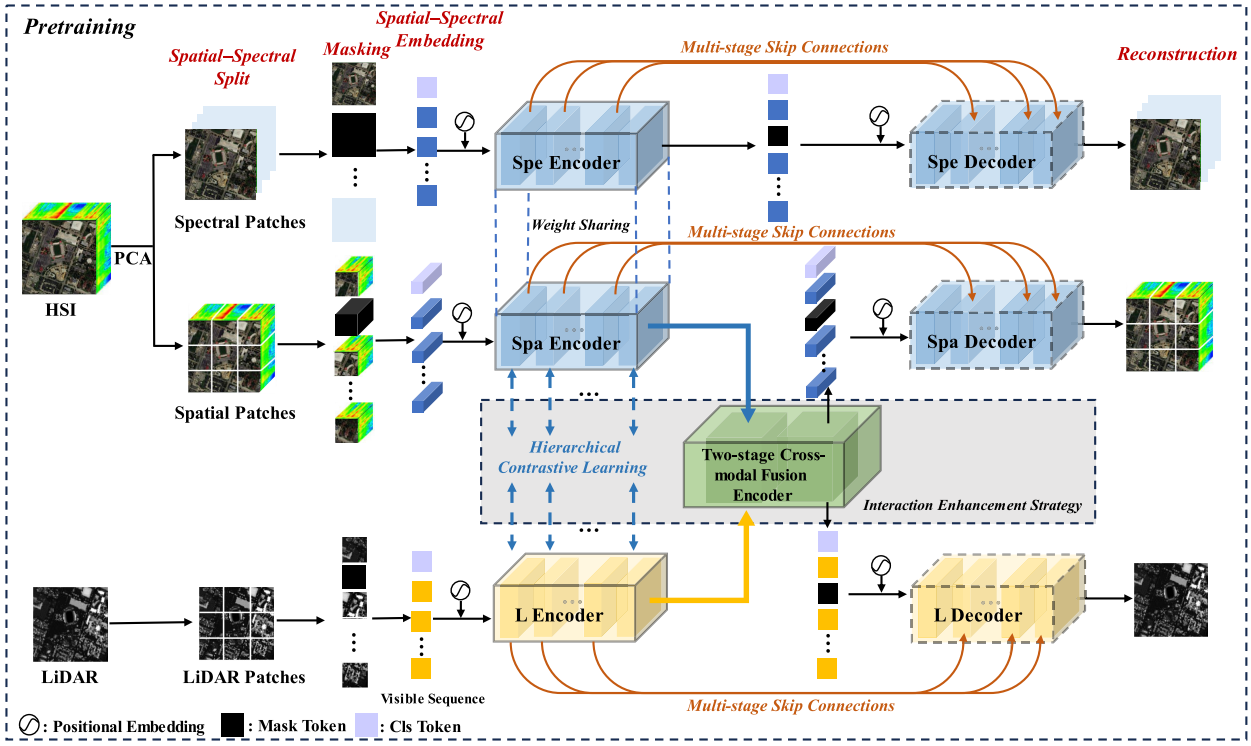


Fig. 1. Overall architecture of the SCIIENet.

is set to 30 in the study. B_2 is the number of channels in the LiDAR/SAR patch. The $\mathbf{X}$ and $\mathbf{X}_{\mathrm{L}}$ split along the spatial tokens, respectively. Additionally, the $\mathbf{X}$ is split along the spectral dimension into a series of spectral bands and embedded into spectral tokens.

Following MAE, we randomly mask a subset for each type of token and obtain the visible (without mask) spectral tokens $\mathbf{X}'_{\mathrm{Spe}} \in \mathbb{R}^{B_1\cdot(1-\rho)\times d}$, visible spatial tokens $\mathbf{X}'_{\mathrm{Spa}} \in \mathbb{R}^{S^2\cdot(1-\rho)\times d}$ and visible LiDAR/SAR tokens $\mathbf{X}'_{\mathrm{L}} \in \mathbb{R}^{S^2\cdot(1-\rho)\times d}$, respectively. Here, ρ is the mask ratio, and d is the feature dimension of each token. Then, $\mathbf{X}'_{\mathrm{Spa}}$ are fed into the Spa branch to reconstruct $\mathbf{X}$ by spatial. $\mathbf{X}'_{\mathrm{Spe}}$ are fed into the Spe branch to reconstruct $\mathbf{X}$ by spectral. $\mathbf{X}'_{\mathrm{L}}$ are fed into the L branch to reconstruct $\mathbf{X}_{\mathrm{L}}$. Each reconstruction branch adopts a transformer encoder–decoder architecture. Specifically, the weights of encoders in Spa and Spe branches are shared to better capture spatial–spectral features from HSIs. The output of each branch is the reconstructed images. The loss for each branch is computed only at the masked positions of the reconstructed and original images by the mean squared error (MSE) loss function. The total reconstruction loss is the sum of the losses from the three branches as (1). Through reconstruction, the complementary spectral features have been pretrained

$$\mathcal{L}_{\mathrm{MAE}} = \mathcal{L}^{\mathrm{Spe}}_{\mathrm{MAE}}(\mathbf{X}, \hat{\mathbf{X}}_{\mathrm{Spe}}) + \mathcal{L}^{\mathrm{Spa}}_{\mathrm{MAE}}(\mathbf{X}, \hat{\mathbf{X}}_{\mathrm{Spa}}) + \mathcal{L}^{\mathrm{L}}_{\mathrm{MAE}}(\mathbf{X}_{\mathrm{L}}, \hat{\mathbf{X}}_{\mathrm{L}}) \tag{1}$$

where $\hat{\mathbf{X}}_m, m \in \{\mathrm{Spe}, \mathrm{Spa}, \mathrm{L}\}$ is the reconstructed images.

Since reconstruction cannot focus on shared information, CL with symmetric information noise contrastive estimation (InfoNCE) loss [54] is employed for the pairs of features from the encoder layer. This method facilitates the alignment of features across modalities. Such consistency implicitly provides additional supervision for these regions, thereby enhancing the performance of feature representation. Specifically, the features after global average pooling encompass both shared and complementary information. If these features become excessively similar, there is a risk of losing complementary information. The reconstruction constraint addresses this issue, thereby aiding in the preservation of most of the complementary information.

C. Interaction Enhancement Strategy

Interactions relying solely on conventional CL have not fully bridged the gap between different modalities or exploited shared and complementary information; therefore, we propose an IES that incorporates HCL and TCFE. On the one hand, the gap between multimodal data is significantly larger than the differences within monomodal data, making it challenging to align features across modalities if only high-level features output from the last encoder layers are considered. In order to address this issue, we introduce HCL to enforce alignment across multiple intermediate features.

Meanwhile, having a similar feature space is beneficial for the subsequent feature fusion process. On the other hand, data from different modalities can exhibit varying degrees of noise and poor quality. In order to minimize adverse effects and effectively integrate beneficial information from different modalities for reconstruction, we propose TCFE, which involves initial fusion followed by separation.

1) Hierarchical CL: In order to better narrow the gap between features of different modalities, HCL aligns the encoding features layer by layer. The encoders of SCIIENet consist of N transformer layers [55], and each transformer layer contains multihead self-attention (MHSA), feed-forward network (FFN), and layer normalization (LN). The specific

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process can be formulated as follows:

$$\mathbf{E}_m^{j'} = \mathrm{MHSA}\left(\mathrm{LN}\left(\mathbf{E}_m^{j-1}\right)\right) + \mathbf{E}_m^{j-1} \tag{2}$$

$$\mathbf{E}_m^{j} = \mathrm{FFN}\left(\mathrm{LN}\left(\mathbf{E}_m^{j'}\right)\right) + \mathbf{E}_m^{j'} \tag{3}$$

where $\mathbf{E}_m^j$ is the encoding features of the j ($j \in \{1 \ldots, N\}$)th layer of the encoder of modality m, and $\mathbf{E}_m^0 = \mathbf{X}'_m$, $m \in \{\mathrm{Spe}, \mathrm{Spa}, \mathrm{L}\}$.

We employ the symmetric InfoNCE loss not only for the encoded features of the final encoder layer but also for the pairs from all preceding encoder layers. Specifically, we first perform global average pooling for the encoding features of Spa and L branches, i.e., $e_{i,m}^j = \mathrm{AvgPool}(\mathbf{E}_{i,m}^j)$, $m \in \{\mathrm{Spa}, \mathrm{L}\}$, where $i \in \{1, \ldots, M\}$ denotes the index of each sample in a batch. We treat sample pairs with the same index across modalities in the batch as positive samples and pairs with different indices as negative samples. The symmetric InfoNCE loss maximizes the cosine similarity for positive pairs while minimizing it for negative pairs. The process at the jth layer of the encoder is shown as follows:

$$\mathcal{L}_{\mathrm{InfoNCE}}\left(\mathbf{e}_{\mathrm{Spa}}^j, \mathbf{e}_{\mathrm{L}}^j\right) = -\frac{1}{2}\left[\mathcal{L}\left(\mathbf{e}_{\mathrm{Spa}}^j, \mathbf{e}_{\mathrm{L}}^j\right) + \mathcal{L}\left(\mathbf{e}_{\mathrm{L}}^j, \mathbf{e}_{\mathrm{Spa}}^j\right)\right] \tag{4}$$

$$\mathcal{L}\left(\mathbf{e}_{\mathrm{Spa}}^j, \mathbf{e}_{\mathrm{L}}^j\right) = -\frac{1}{M}\sum_{i=1}^{M} \log \frac{\exp\left(\mathrm{sim}\left(\mathbf{e}_{i,\mathrm{Spa}}^j, \mathbf{e}_{i,\mathrm{L}}^j\right)/\tau\right)}{\sum_{q=1}^{M} \exp\left(\mathrm{sim}\left(\mathbf{e}_{i,\mathrm{Spa}}^j, \mathbf{e}_{q,\mathrm{L}}^j\right)/\tau\right)} \tag{5}$$

where $\mathrm{sim}(\cdot)$ denotes the cosine similarity function and τ is the temperature factor.

The overall contrastive loss is the sum of the losses across all encoder layers

$$\mathcal{L}_{\mathrm{HCL}} = \sum_{j=1}^{N} \mathcal{L}_{\mathrm{InfoNCE}}\left(\mathbf{e}_{\mathrm{Spa}}^j, \mathbf{e}_{\mathrm{L}}^j\right). \tag{6}$$

Finally, the overall pretraining loss of the SCIIENet is the sum of the reconstruction and contrastive loss

$$\mathcal{L} = \mathcal{L}_{\mathrm{MAE}} + \lambda \mathcal{L}_{\mathrm{HCL}} \tag{7}$$

where λ is a weighting factor that balances the reconstruction loss with the contrastive loss.

2) Two-Stage Cross-Modal Fusion Encoder: We designed the TCFE module to fuse multimodal features during the pretraining stage, as shown in Fig. 2. Because the fused features can be enhanced by shared and complementary information, they are less affected by poorly captured data and contain more complete information compared to features extracted from any individual modality. Exchanging information from the fused features is more effective compared to single-stage cross-attention used in previous studies [56], [57] for exchanging information from the other modalities. Motivated by this, TCFE first employs the encoding features as agents to exchange information with encoding features from another branch and obtains the fused features. Subsequently, TCFE uses the encoding features to exchange information with the fused features and projects them back to their respective branches.

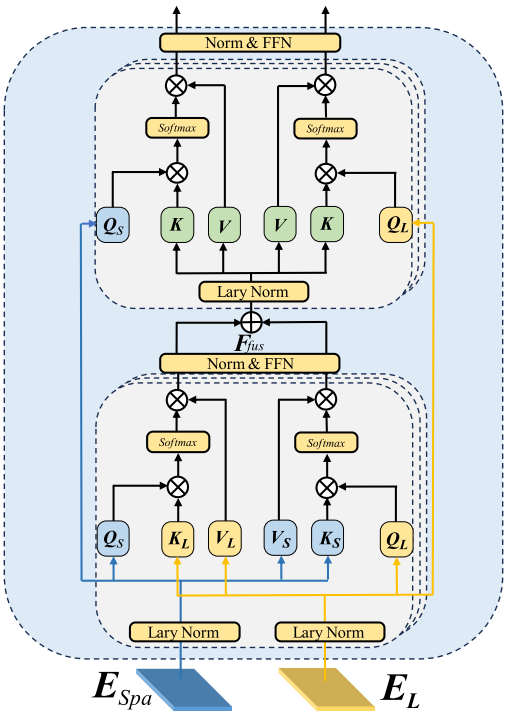


Fig. 2. Illustration of a TCFE.

The cross-attention mechanism of the first stage of TCFE is shown in (8) and (9), taking the Spa branch as an example. Multihead cross-attention (MHCA), a variant of MHSA, is employed for the cross-attention operation. The key distinction in MHCA is that the query (**Q**) is derived from one feature, whereas the key (**K**) and value (**V**) are obtained from another feature. In this context, the encoding features of the Spa branch $\mathbf{E}_{\mathrm{Spa}}^N$ are used to generate **Q**, whereas the encoding features of the L branch $\mathbf{E}_L^N$ are employed to produce **K** and **V**. Following the MHCA, the FFN is applied, and the outputs of the FFN from both branches are summed to obtain the fused feature $\mathbf{F}_{\mathrm{fus}}$ as (10):

$$\mathbf{F}_{\mathrm{Spa}}^{1'} = \mathrm{MHCA}\left(\mathrm{LN}\left(\mathbf{E}_{\mathrm{Spa}}^N\right), \mathrm{LN}\left(\mathbf{E}_{\mathrm{L}}^N\right), \mathrm{LN}\left(\mathbf{E}_{\mathrm{L}}^N\right)\right) + \mathbf{E}_{\mathrm{Spa}}^N \tag{8}$$

$$\mathbf{F}_{\mathrm{Spa}}^{1} = \mathrm{FFN}\left(\mathrm{LN}\left(\mathbf{F}_{\mathrm{Spa}}^{1'}\right)\right) + \mathbf{F}_{\mathrm{Spa}}^{1'} \tag{9}$$

$$\mathbf{F}_{\mathrm{fus}} = \mathbf{F}_{\mathrm{Spa}}^{1} \oplus \mathbf{F}_{\mathrm{L}}^{1} \tag{10}$$

where $\oplus$ is a cascade operation.

The cross-attention mechanism in the second stage of the TCFE is illustrated in (11) and (12), again using the Spa branch as an example. The encoding features of the Spa branch $\mathbf{E}_{\mathrm{Spa}}^N$ are used to generate the **Q**. The fused features contain rich information that enables the agents to acquire more beneficial insights, which are used to generate the **K** and **V**. The cross-modal features output from the TCFE are subsequently fed into the decoders for reconstruction

$$\mathbf{F}_{\mathrm{Spa}}^{2'} = \mathrm{MHCA}\left(\mathrm{LN}\left(\mathbf{E}_{\mathrm{Spa}}^N\right), \mathrm{LN}(\mathbf{F}_{\mathrm{fus}}), \mathrm{LN}(\mathbf{F}_{\mathrm{fus}})\right) + \mathbf{E}_{\mathrm{Spa}}^N \tag{11}$$

$$\mathbf{F}_{\mathrm{Spa}}^{2} = \mathrm{FFN}\left(\mathrm{LN}\left(\mathbf{F}_{\mathrm{Spa}}^{2'}\right)\right) + \mathbf{F}_{\mathrm{Spa}}^{2'}. \tag{12}$$

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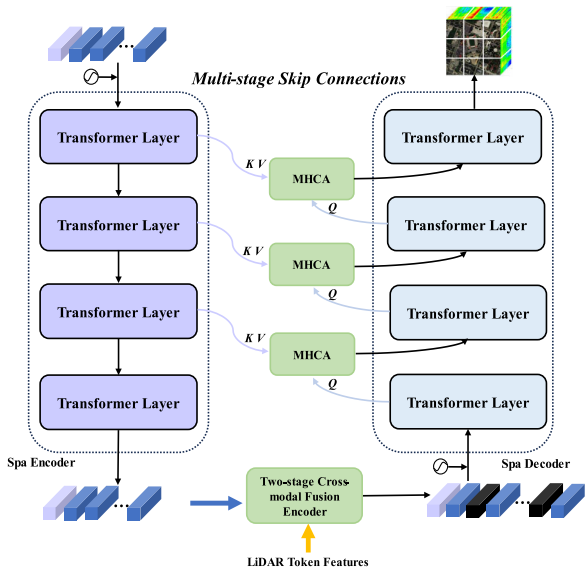


Fig. 3. Illustration of MSCs.

D. Multistage Skip Connections

Better reconstruction leads to the preservation of more information, which benefits downstream tasks. Low-level detail features in the middle layers of the encoder are crucial for guiding representation learning to achieve better reconstruction [58]. Compared to conventional MAE that reconstructs images using only the last layer of the encoder, the MSCs are, therefore, proposed to integrate intermediate layer features into the decoding process for each branch.

The decoder is composed of N transformer layers. MHCA is employed to connect the encoder and decoder in MSCs. In the MSCs, the encoding features from the encoder layers serve as the **K** and **V**, while the output of the decoder layers acts as the **Q**. The output of the MSCs subsequently fed into the subsequent layers of the decoder. Through the attention-based mechanism, the MSCs can better extract useful information from the encoders to enhance the performance of the decoders. The MSCs are illustrated in Fig. 3 (using the Spa branch as an example) and are defined by the following:

$$\mathbf{D}_m^k = \mathrm{MHCA}\left(\mathrm{LN}\left(\mathbf{D}_m^{k-1}\right), \mathrm{LN}\left(\mathbf{E}_m^j\right), \mathrm{LN}\left(\mathbf{E}_m^j\right)\right) + \mathbf{D}_m^{k-1} \quad (13)$$

where $\mathbf{D}_m^k$ is the decoding features of the k ($k \in \{2, \ldots, N\}$)th layer of the decoder of modality m, $m \in \{\mathrm{Spe}, \mathrm{Spa}, \mathrm{L}\}$. Specifically, $k + j = N + 1$. $\mathbf{D}_m^1$ is the output of the first transformer layer of the decoder, using $\mathbf{F}_{\mathrm{Spa}}^2$ and $\mathbf{F}_{\mathrm{L}}^2$ as input for the Spa and L branches respectively, while $\mathbf{E}_{\mathrm{Spe}}^N$ serves as input for the spe branch. Finally, $\mathbf{D}_m^N$ is passed through the corresponding linear projection layer to obtain the reconstructed data $\hat{\mathbf{X}}_m$.

E. Fine-Tuning

Since the encoders have been well-trained to obtain shared and complementary information during the pretraining stage, all the encoders with their weights are used for fine-tuning. In order to fuse features from three encoder middle layers, we employ adaptive multistage feature fusion (AMFF) [19] as (16) and (17), which assigns trainable weights to the feature of each level and subsequently sums them. These features output from AMFF are then integrated with the cross-modal features to obtain an overall feature representation. The process is shown as follows:

$$\mathbf{F}_{\mathrm{HSI}}^{\mathrm{AMFF}} = \sum_{j=1}^{N} \alpha_{\mathrm{HSI}}^j \left(\lambda_{\mathrm{Spe}}^j \mathbf{e}_{\mathrm{Spe}}^j + \lambda_{\mathrm{Spa}}^j \mathbf{e}_{\mathrm{Spa}}^j\right) \quad (14)$$

$$\mathbf{F}_{\mathrm{L}}^{\mathrm{AMFF}} = \sum_{j=1}^{N} \alpha_{\mathrm{L}}^j \mathbf{e}_{\mathrm{L}}^j \quad (15)$$

where λ_{Spe}^j, λ_{Spa}^j, α_m^j represents the trainable weights, $\sum_{j=1}^N \alpha_m^j = 1$, $m \in \{\mathrm{HSI}, \mathrm{L}\}$, respectively,

$$\mathbf{E}_{\mathrm{HSI}} = \mathbf{F}_{\mathrm{Spa}}^2 + \mathbf{F}_{\mathrm{HSI}}^{\mathrm{AMFF}}, \quad \mathbf{E}_{\mathrm{L}} = \mathbf{F}_{\mathrm{L}}^2 + \mathbf{F}_{\mathrm{L}}^{\mathrm{AMFF}} \quad (16)$$

$$\mathbf{E} = \mathrm{Concat}(\mathbf{E}_{\mathrm{HSI}} \mathbf{E}_{\mathrm{L}}). \quad (17)$$

Finally, the **E** is fed to fully connected layers for classification. The cross-entropy loss function is used as the classification loss.

IV. Experimental Results and Analysis

In order to validate the effectiveness of the proposed method, we conducted comprehensive experiments on four multimodal remote sensing datasets: Houston 2013, Augsburg, Berlin, and MUUFL. These experiments included both quantitative and qualitative assessments.

A. Dataset Description

1) Houston 2013 Dataset: The dataset is the HSI-LiDAR dataset acquired by the ITRES CASI-1500 imaging sensor and covers the University of Houston campus and adjacent areas. A digital surface model (DSM) image was generated using LiDAR data. Both the spatial size of HIS and LiDAR data is 349 × 1905 with a spatial resolution of 2.5 m. The HSI has 144 bands with a wavelength range from 0.38 to 1.05 μm. In all, the dataset consists of 15 029 feature samples and 15 land cover types.

2) MUUFL Dataset: The dataset was captured by the ITRES CASI-1500 imaging sensor at the University of Southern Mississippi's Gulf Coast campus in November 2010. Because of the influence of imaging noise, the first four and last four bands of HSI were removed, and 64 bands were ultimately used. The invalid area on the right side of the original image was removed, leaving 325 × 220 pixels preserved, with a spatial resolution of 1 m. Coregistered LiDAR data with the same resolution are available. The dataset includes a total of 53 687 feature samples and 11 land cover types.

3) Augsburg Dataset: The dataset consists of HSI and SAR data, which were captured by the HySpex and Sentinel-1 sensors, respectively. The total size of the dataset is 332 × 485 with a spatial resolution of 30 m. The HSI contains 180 bands in the wavelength range of 0.4–2.5 μm, and the SAR image has four features derived from polarization decomposition. In total, the dataset includes 78 294 feature samples and seven land cover types.

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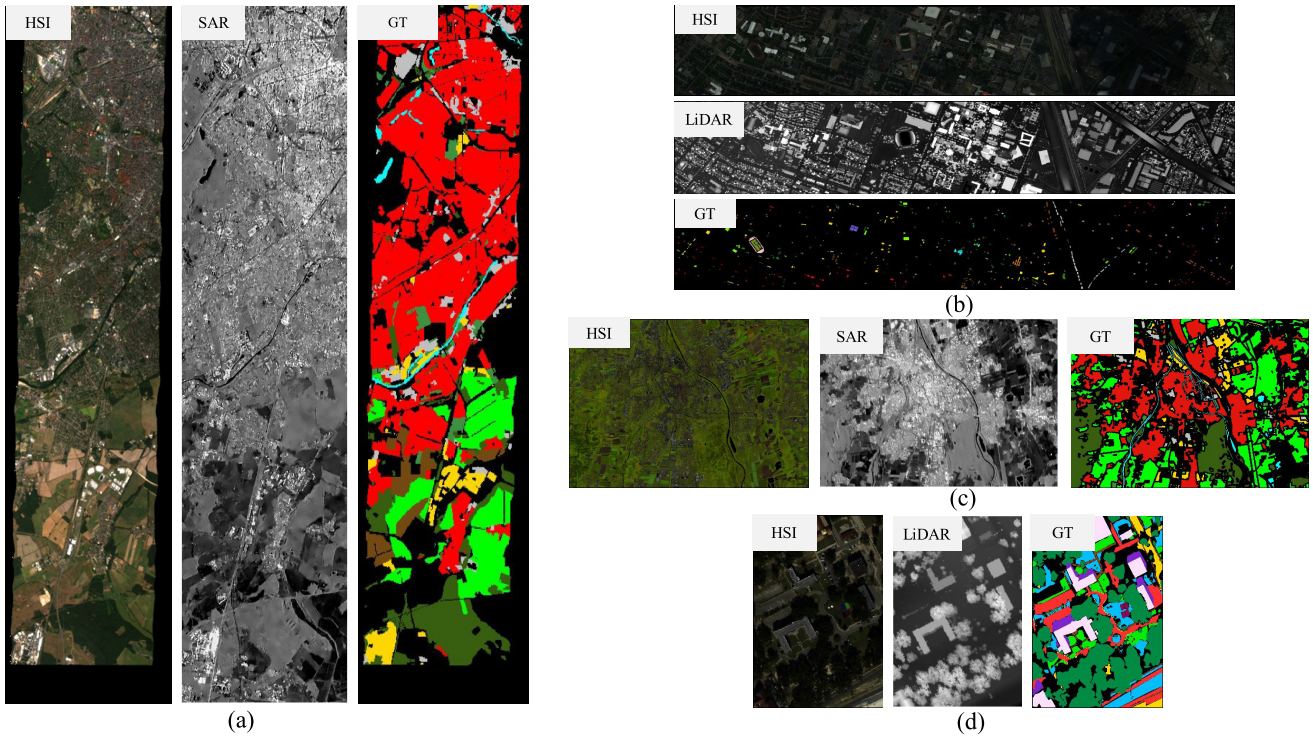


Fig. 4. Visualization of the four datasets. (a) Berlin dataset. (b) Houston 2013 dataset. (c) Augsburg dataset. (d) MUUFL dataset.

4) Berlin Dataset: The dataset covers the geomorphological composition of the urban and rural areas around Berlin. The HSI is EnMAP data synthesized from the onboard HyMap data with a spatial resolution of 30 m. The HSI contains 244 bands in the wavelength range of 0.4–2.5 μm. The two-channel PolSAR image was obtained from the European Space Agency's Sentinel-1 dual polarization (VV-VH) single view complex (SLC) product, measuring 1723 $\times$ 476 with a spatial resolution of 13 m. The HSI was processed using a nearest-neighbor interpolation algorithm to give it the same spatial resolution as the SAR image. The dataset consists of 464 671 feature samples and eight land cover types.

The visualized images of the four datasets are shown in Fig. 4, including the false color image of HSI, the gray-scale image of LiDAR, and the SAR image. The category names, number of training samples, and test samples for the four datasets are shown in Tables I–IV; especially, all the data from each dataset is employed to pretrain the model for that dataset.

TABLE I
NUMBER OF TRAINING TEST SAMPLES FOR THE HOUSTON 2013 DATASET

No.	Class Name	Training	Testing
1	Healthy grass	10	1241
2	Stressed grass	10	1244
3	Synthetic grass	10	687
4	Trees	10	1234
5	Soil	10	1232
6	Water	10	315
7	Residential	10	1258
8	Commercial	10	1234
9	Road	10	1242
10	Highway	10	1217
11	Railway	10	1225
12	Parking lot1	10	1223
13	Parking lot2	10	459
14	Tennis court	10	418
15	Running track	10	650
	Total	150	14879

B. Experimental Settings

The experiments for the proposed method and other comparative methods were conducted on the PyTorch platform in Python 3.7. The Adam optimizer was chosen to optimize the network. During the pretraining phase, the learning rate was set to 1e-4, with a batch size of 128 and 50 training epochs. For the fine-tuning stage, the learning rate was adjusted to 5e-4, using a batch size of 16 and 100 training epochs. Additionally, a weight decay factor of 5e-2 was applied throughout the training process. The classification performance was evaluated using overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (κ).

C. Parameter Settings

1) Patch Size: The size of the input patches affects the model's ability to capture global information, and thus, impacts classification performance. We tested five different input patch sizes: 3 $\times$ 3, 5 $\times$ 5, 7 $\times$ 7, 9 $\times$ 9, and 11 $\times$ 11. As shown in Fig. 5(a), the optimal classification performance for the Houston 2013, Augsburg, Berlin, and MUUFL datasets is achieved with patch sizes of 9 $\times$ 9, 7 $\times$ 7, 7 $\times$ 7, and 11 $\times$ 11, respectively. The size of the selected patch

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TABLE II
NUMBER OF TRAINING TEST SAMPLES FOR THE MUUFL DATASET

No.	Class Name	Training	Testing
1	Trees	10	23236
2	Mostly grass	10	4260
3	Mixed ground surface	10	6872
4	Dirt and sand	10	1816
5	Road	10	6677
6	Water	10	456
7	Building shadow	10	2223
8	Building	10	6230
9	Sidewalk	10	1375
10	Yellow curb	10	173
11	Cloth panels	10	259
	Total	110	53577

TABLE III
NUMBER OF TRAINING TEST SAMPLES FOR THE AUGSBURG DATASET

No.	Class Name	Training	Testing
1	Forest	10	13497
2	Residential area	10	30319
3	Industrial area	10	3841
4	Low plants	10	26847
5	Allotment	10	565
6	Commercial area	10	1635
7	Water	10	1520
	Total	70	78224

TABLE IV
NUMBER OF TRAINING TEST SAMPLES FOR THE BERLIN DATASET

No.	Class Name	Training	Testing
1	Forest	50	54904
2	Residential area	50	268592
3	Industrial area	50	19516
4	Low plants	50	59232
5	Soil	50	17376
6	Allotment	50	13255
7	Commercial area	50	24774
8	Water	50	6622
	Total	400	464271

correlates with the spatial resolution of the dataset, primarily due to the close relationship between spatial resolution and category information. Increasing the patch size at low spatial resolution introduces irrelevant information, thereby degrading classification performance.

2) Mask Ratio: Different masking ratios affect the robustness of the pretraining feature representations and help reduce the complexity of the model. As shown in Fig. 5(b), the classification accuracy for different datasets increases with the masking ratio. When the masking ratio exceeds a certain threshold, fewer features can, however, be extracted by the model, resulting in information loss and decreased classification accuracy. In this context, the masking rate for MUUFL is set to 80%, while OA reaches its highest value when the masking rate is set to 70% for the other datasets. This is because the spatial resolution of the MUUFL dataset is the lowest, and a higher masking rate is necessary to break the local relationship between pixels.

3) Head Number: The number of attention heads influences the expressiveness of the model and the allocation of attention, which in turn affects classification performance. The impact of different numbers of attention heads on the classification performance of our model across the four datasets, using the candidate sets $\{2, 4, 8, 16, 32\}$, is shown in Fig. 5(c). The results indicate that an insufficient number of attention heads diminishes classification performance, while increasing the number of attention heads beyond eight leads to stable classification performance. The model achieves the best classification performance on the four datasets when the number of attention heads is set to 8.

4) Token Dimension: The different token dimensions of the hidden layer will impact the expressiveness of the model. We comprehensively selected the optimal token dimension for each dataset from the candidate dimensions $\{32, 64, 128, 256, 512\}$. As shown in Fig. 6(a), the model achieves the highest OA values across the four datasets when the token dimension is set to 256.

5) Model Depth: The model depth of the network corresponds to N layers of encoder and decoder, which primarily refers to the expressiveness and complexity of the model. We selected the optimal N for each dataset from the candidate dimensions $\{2, 3, 4, 5, 6\}$. The OA of models with different N across the four datasets is presented in Fig. 6(b). The results indicate that the proposed network achieves the best classification results when the encoder depth is set to 3.

6) Loss Weight: Different values of the loss weight coefficient λ affect the rate of reconstruction and contrastive loss. Fig. 6(c) illustrates the variation of OA when λ is set to $\{0.1, 0.01, 0.005, 0.001, 0.0005, 0.0001\}$ for each of the four datasets, respectively. A larger λ value would improve the risk of omitting complementary information, potentially detrimental to performance; therefore, setting λ to 0.1 resulted in the poorest classification performance. CL is, moreover, more sensitive to dataset size than reconstruction, necessitating a higher λ value to enhance its efficacy, especially for smaller datasets; therefore, λ for Houston 2013 is set to 0.001, as the dataset has the smallest sample size, while λ is set to 0.0005 for the other datasets.

D. Classification Results and Analysis

In order to evaluate the effectiveness of the proposed model, we compared SCIIENet with nine state-of-the-art methods: FusAtNet [59], multimodal deep learning (MDL) [60], EndNet [61], spatial-spectral enhancement network (S^2ENet) [62], ExViT [40], hierarchical CNN and transformer network (HCTNet) [63], Fus-MAE [38], nearest neighbor-based contrastive learning network (NNCNet) [33] and spatial–spectral masked autoencoder (SS–MAE) [37]. Among them, Fus-MAE, NNCNet, and SS-MAE are self-supervised methods, while the others are supervised methods.

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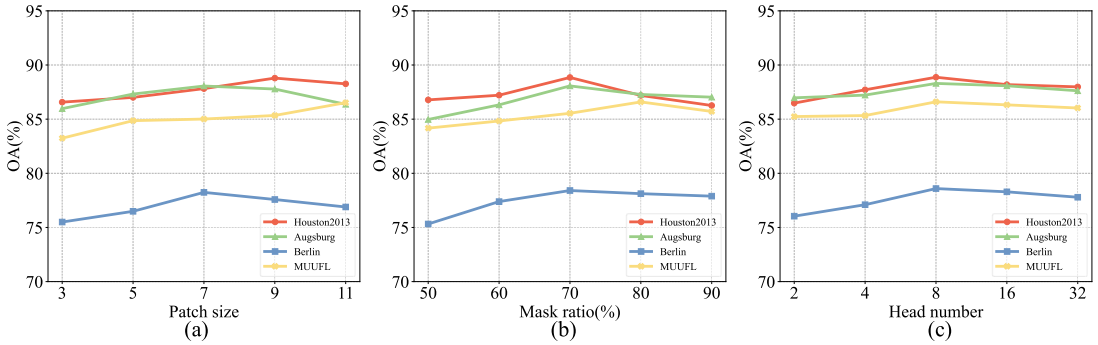


Fig. 5. Effect of different parameters on OA. (a) Patch size. (b) Mask ratio. (c) Head number.

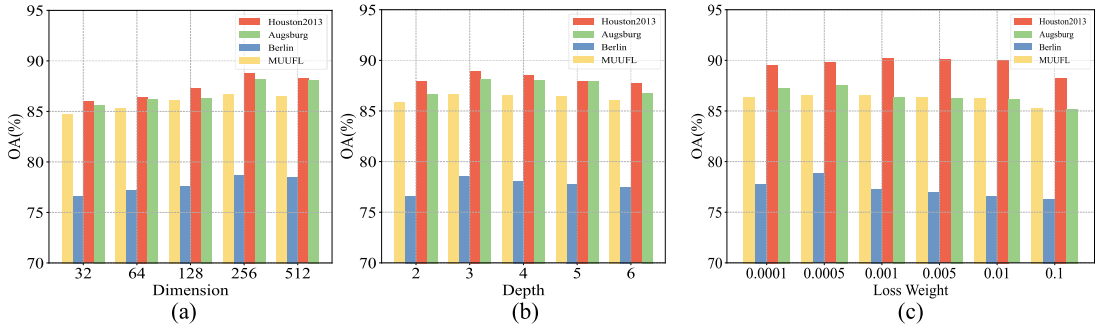


Fig. 6. Effect of different parameters on OA. (a) Token dimension. (b) Model depth. (c) Loss weight.

TABLE V

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE HOUSTON 2013 DATASET (THE OPTIMAL RESULT IS HIGHLIGHTED IN BOLD FONT)

Class	Supervised methods						Self-supervised methods			
	FusAtNet	MDL	EndNet	S^{2}ENet	ExViT	HCTnet	Fus-MAE	NNCNet	SS-MAE	Proposed
Healthy grass	90.47±3.88	96.55±1.48	88.79±6.01	**97.81±0.31**	96.38±1.74	93.52±1.21	94.82±0.71	91.45±2.38	97.53±0.21	97.55±0.63
Stressed grass	90.14±3.20	89.06±4.55	96.09±2.27	92.05±3.89	96.73±2.62	92.53±3.06	98.71±0.39	**98.84±0.21**	98.64±0.49	94.56±2.50
Synthetic grass	92.14±2.72	97.23±0.40	**99.94±0.11**	98.74±1.23	96.50±0.49	96.91±2.45	99.76±0.07	99.71±0.13	99.82±0.57	99.68±0.34
Trees	82.60±4.51	96.96±0.06	95.13±0.38	98.82±0.61	94.42±0.77	94.78±3.74	**99.83±0.12**	99.41±0.06	99.17±0.21	99.56±0.76
Soil	87.17±3.05	99.26±0.41	96.64±2.04	99.93±0.07	98.30±0.13	98.70±1.34	**100.00**	99.96±0.03	**100.00**	**100.00**
Water	76.19±19.41	90.60±0.82	97.08±1.14	96.51±3.11	95.61±1.05	97.84±0.17	93.25±0.15	**100.00**	97.45±0.52	97.33±2.17
Residential	86.07±7.44	87.91±0.70	85.72±5.22	81.55±2.98	88.07±2.74	95.86±0.23	91.02±1.95	95.16±1.53	**99.04±0.24**	94.30±0.73
Commercial	60.98±2.29	55.73±1.86	59.25±2.52	60.21±1.71	59.09±0.40	56.37±4.03	59.90±1.49	57.79±2.98	59.37±2.34	**71.40±5.44**
Road	79.46±7.39	65.20±8.02	72.19±1.91	82.90±4.54	78.29±2.44	**86.06±1.01**	82.57±0.97	81.11±2.08	73.23±0.94	81.81±6.59
Highway	49.11±17.10	59.42±4.66	50.32±6.73	54.76±1.98	64.33±0.20	64.53±5.50	89.36±0.55	72.21±10.07	90.47±4.71	**92.84±1.27**
Railway	74.68±2.66	73.19±8.06	86.53±4.62	91.69±0.71	88.24±4.35	86.97±2.01	90.95±0.56	**97.02±3.65**	95.91±0.89	93.57±2.25
Parking lot1	40.11±0.87	45.65±3.86	42.38±4.55	45.65±7.88	63.90±2.15	44.28±2.37	42.38±0.84	60.60±13.67	43.13±4.47	**69.04±2.94**
Parking lot2	81.74±4.75	86.44±7.71	54.16±3.05	96.38±1.97	94.98±1.72	**99.17±0.63**	97.99±0.25	96.21±1.39	93.94±1.8	98.43±1.28
Tennis court	99.52±0.30	**100.00**	98.66±0.73	**100.00**	**100.00**	**100.00**	99.61±0.11	**100.00**	**100.00**	**100.00**
Running track	86.24±4.04	99.10±0.33	98.49±0.11	99.29±0.80	**100.00**	**100.00**	99.78±0.07	**100.00**	**100.00**	**100.00**
OA(%)	74.19±1.74	80.11±1.68	80.49±2.42	84.90±0.49	85.34±0.85	86.31±0.65	87.32±0.37	88.75±0.19	88.77±0.16	**90.18±0.90**
AA(%)	77.57±2.60	82.82±1.66	81.44±1.16	86.76±0.60	87.66±0.58	87.17±1.74	89.35±0.29	90.07±0.44	90.06±0.14	**92.27±1.37**
$\kappa\times100$	72.16±1.91	78.50±1.83	78.03±1.46	84.12±1.04	84.17±0.92	85.95±1.36	86.30±0.40	87.78±0.23	87.76±0.17	**89.38±0.97**

In particular, FusAtNet emphasizes spectral–spatial features for MRSI classification by using self-attention and cross-attention mechanisms. MDL introduces a weight-sharing dual-branch DL framework with different-stage fusion. EndNet integrates multimodal information by a reconstruction strategy through an encoder–decoder network architecture. S^{2}ENet is a spatial–spectral enhancement network designed to simultaneously improve spatial and spectral feature representations. ExViT employs parallel branches of position-shared transformer extended with separable convolution modules and a cross-modality attention module to fuse tokenized embeddings. HCTNet combines hierarchical CNNs and transformer networks to extract features from multisource data and subsequently fuses these features using a cross-token attention

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TABLE VI
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE AUGSBURG DATASET (THE OPTIMAL RESULT IS HIGHLIGHTED IN BOLD FONT)

Class	Supervised methods						Self-supervised methods			
	FusAtNet	MDL	EndNet	S^2ENet	ExViT	HCTnet	Fus-MAE	NNCNet	SS-MAE	Proposed
Forest	86.10±2.19	91.24±2.75	93.52±1.27	93.28±0.13	91.51±4.02	93.26±4.54	95.70±3.41	96.83±0.47	96.44±0.34	**98.56±0.79**
Residential area	74.25±1.53	77.24±3.04	71.29±1.65	77.87±2.71	78.02±5.17	79.92±6.28	76.26±0.82	78.92±2.88	85.01±0.62	**87.57±3.78**
Industrial area	36.26±2.27	53.49±10.02	40.25±3.75	46.84±2.89	**68.88±1.83**	48.77±0.43	47.62±5.52	46.13±1.90	52.69±4.15	56.05±7.93
Low plants	62.95±5.54	55.53±2.07	71.61±1.14	72.62±3.54	70.77±0.52	71.86±2.11	79.98±3.39	78.39±3.65	82.14±2.01	**91.03±3.24**
Allotment	63.76±0.62	80.92±2.86	80.07±1.45	90.72±0.14	85.32±4.31	83.21±7.98	95.25±2.98	91.92±0.28	94.59±0.21	**96.67±2.74**
Commercial area	48.52±0.67	32.07±10.96	37.79±11.80	45.67±1.51	38.37±5.12	**55.97±0.69**	45.08±3.52	44.31±5.87	52.63±5.69	43.62±6.24
Water	58.46±3.47	62.90±2.91	61.81±4.91	67.08±0.91	55.61±1.25	66.49±4.94	63.90±6.52	**80.27±1.41**	66.90±1.12	64.93±1.03
OA(%)	69.64±1.91	69.55±0.99	74.05±0.90	76.41±0.30	75.71±0.86	76.94±0.52	79.53±1.45	79.61±0.52	83.45±0.58	**87.57±0.85**
AA(%)	62.27±1.59	64.33±1.39	63.50±1.28	70.58±0.34	69.51±0.42	71.35±1.88	71.97±0.78	73.82±0.73	75.77±0.59	**76.21±0.75**
$\kappa\times100$	59.32±2.49	59.80±1.10	64.66±0.97	68.54±0.37	67.84±1.58	68.99±0.44	71.35±0.22	72.75±0.63	77.29±0.76	**82.77±1.06**

TABLE VII
CLASSIFICATION PERFORMANCE WAS OBTAINED BY DIFFERENT METHODS FOR THE BERLIN DATASET (THE OPTIMAL RESULT IS HIGHLIGHTED IN BOLD FONT)

Class	Supervised methods						Self-supervised methods			
	FusAtNet	MDL	EndNet	S^2ENet	ExViT	HCTnet	Fus-MAE	NNCNet	SS-MAE	Proposed
Forest	66.08±6.00	81.74±6.18	82.59±1.67	85.78±0.44	83.31±0.68	83.13±1.33	80.51±0.62	78.14±4.84	81.01±2.97	**86.17±1.84**
Residential area	65.62±1.73	57.44±0.88	44.69±0.88	56.47±1.87	60.36±1.34	61.07±0.73	64.04±2.04	76.83±1.98	70.05±2.25	**83.28±0.49**
Industrial area	61.44±2.10	61.68±2.10	48.15±1.11	60.76±1.09	64.24±1.42	61.91±1.14	59.19±0.97	57.56±4.85	**65.76±1.59**	63.70±4.26
Low plants	61.86±3.29	73.05±0.62	76.86±1.86	84.87±1.05	83.97±0.80	84.11±1.28	83.29±2.30	**87.62±1.63**	77.51±2.69	79.49±5.55
Soil	88.32±0.74	93.27±0.38	89.29±0.39	92.72±0.74	93.58±0.14	93.44 ±1.30	90.53±2.43	73.89±0.48	**95.26±0.12**	91.81±1.28
Allotment	55.72±4.43	76.21±0.15	78.37±0.57	84.61±0.56	77.48±1.50	79.42±0.35	85.11±0.68	**86.49±0.73**	78.37±3.46	72.02±4.77
Commercial area	45.92±2.66	43.50±0.35	38.62±1.25	47.38±3.73	55.55±2.29	53.82±1.16	59.37±1.93	**62.71±3.45**	53.81±0.59	52.70±3.89
Water	85.16±3.11	93.62±0.94	73.79±2.61	95.64±0.55	95.59±0.62	94.46±1.47	**97.66±0.19**	90.97±0.72	93.16±0.12	84.99±4.01
OA(%)	64.72±0.28	64.41±0.63	63.82±0.35	65.98±1.02	68.18±0.18	69.42±0.73	70.29±1.12	74.29±0.36	73.36±0.68	**78.82±0.11**
AA(%)	67.00±1.04	73.12±0.37	66.05±0.29	76.06±0.38	77.41±0.38	76.42±0.80	77.46±0.29	77.25±0.04	76.88±0.73	**77.76±0.81**
$\kappa\times100$	49.88±0.05	52.86±0.32	42.26±0.25	54.98±0.97	55.66±1.73	57.25±1.24	59.51±1.21	65.87±1.57	61.72±0.41	**68.25±0.32**

fusion encoder. Fus-MAE employs cross-attention for early and feature-level fusion to obtain fused features specifically for MAEs. NNCNet applies CL for hyperspectral and LiDAR data classification through a nearest-neighbor-based data augmentation technique. SS-MAE learns generic feature representations in multimodal data by randomly masking and reconstructing patches that branch spatially and spectrally. All methods were implemented using their proposed codes, employing the same training and test samples as in our fine-tuning phase.

1) Quantitative Classification Results and Analysis: In Tables V–VIII, the OA, AA, κ, and per-class classification results for each model on the Houston 2013, Augsburg, Berlin, and MUUFL datasets are given separately. For the supervised methods, FusAtNet, MDL, EndNet, and S^2ENet are based on CNNs. The MDL interacts with information through weight sharing, while FusAtNet employs a CNNs-based cross-attention mechanism that leverages LiDAR/SAR features to weight HSI features. This cross-attention mechanism allows for a focus on the most informative features, resulting in better classification performance for FusAtNet compared to MDL. Meanwhile, EndNet uses a reconstruction strategy that enhances the activation of cross-modal neurons and leverages the encoder–decoder architecture to achieve a more compact fusion of multimodal data, resulting in improved performance.

As highlighted in S^2ENet, the aforementioned methods fail to leverage the strengths of one modality to complement another, leading to significantly poorer performance compared to S^2ENet. On the other hand, ExViT and HCTNet are based on transformer architectures. Both methods outperform the CNN-based models, indicating that the multihead attention mechanism can effectively capture beneficial features for fusion. Compared to the MHSA used in ExVIT, HCTNet employs a cross-token attention fusion encoder that extracts features from both modalities by exchanging $\mathbf{Q}$ vectors, facilitating better fusion of complementary information, and thus, achieving superior results. These findings suggest that the cross-attention mechanism, full consideration of complementary information, and reconstruction strategy contribute to more effective feature fusion. All these beneficial aspects have been integrated into our method, resulting in improved performance.

Compared to supervised methods, self-supervised approaches yielded better results. The limitation of labeled samples for training led to overfitting in supervised methods, which critically affected classification performance. Among

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TABLE VIII
CLASSIFICATION PERFORMANCE WAS OBTAINED BY DIFFERENT METHODS FOR THE MUUFL DATASET (THE OPTIMAL RESULT IS HIGHLIGHTED IN BOLD FONT)

Class	Supervised methods						Self-supervised methods			
	FusAtNet	MDL	EndNet	S^2ENet	ExViT	HCTnet	Fus-MAE	NNCNet	SS-MAE	Proposed
Trees	81.01±0.87	85.27±3.94	83.19±0.67	85.76±2.26	87.54±3.85	92.34±0.83	92.11±0.35	91.77±1.01	**93.35±0.26**	92.57±0.71
Mostly grass	68.49±2.85	64.18±7.79	**80.72±4.28**	70.13±3.41	54.13±3.67	66.19±1.81	76.82±4.01	75.25±2.87	72.92±0.86	79.29±1.73
Mixed ground surface	34.07±2.05	48.13±4.40	53.88±2.11	56.05±1.83	57.77±2.25	55.93±0.20	56.45±4.33	**73.21±2.06**	61.31±2.02	68.88±2.93
Dirt and sand	38.60±1.29	40.33±0.47	55.95±1.21	55.99±1.44	64.91±2.97	72.98±1.71	73.40±0.17	66.29±7.48	65.33±2.54	**81.37±0.98**
Road	71.75±1.50	73.75±0.61	**90.14±0.06**	78.44±1.30	79.16±0.94	84.64±1.54	86.01±1.25	82.12±2.02	83.87±0.66	85.41±0.79
Water	81.06±1.16	71.40±13.85	95.78±0.16	78.15±0.32	80.96±5.33	90.80±1.45	96.97±1.22	98.28±0.69	96.01±0.35	**99.82±0.16**
Building shadow	75.63±1.35	71.71±8.38	81.61±7.81	78.84±0.44	75.26±0.81	73.69±4.30	79.35±1.62	80.92±1.85	83.02±0.51	**84.47±0.39**
Building	85.96±1.28	84.91±2.15	90.89±0.27	83.41±1.54	93.22±1.62	89.24±5.04	95.45±1.41	**96.42±0.52**	94.49±0.34	95.72±1.38
Sidewalk	44.93±3.05	50.38±5.74	52.04±3.46	47.12±2.87	41.89±4.94	52.59±0.88	55.34±2.16	52.24±4.62	52.52±1.08	**58.82±3.61**
Yellow curb	52.11±1.65	55.72±1.29	71.05±0.87	63.58±1.26	54.33±1.89	66.96±0.91	75.45±0.76	76.99±1.52	73.63±1.49	**78.26±3.98**
Cloth panels	82.87±2.52	88.03±6.18	75.44±2.12	99.07±0.19	93.66±4.99	80.58±2.19	92.27±0.24	98.14±0.89	**99.30±0.15**	98.07±0.77
OA(%)	69.46±2.03	74.21±0.82	79.62±0.26	77.16±0.40	80.84±0.61	81.75±0.42	83.79±0.35	84.54±0.93	84.33±0.34	**86.61±0.19**
AA(%)	65.25±0.77	66.71±2.17	76.54±2.03	72.41±0.49	71.97±1.21	75.35±0.34	79.04±0.26	81.08±0.25	79.62±0.24	**83.88±0.38**
$\kappa\times100$	64.73±0.54	66.14±1.21	73.95±0.32	70.87±0.35	73.89±0.65	76.04±0.62	78.84±0.48	79.78±1.18	79.23±0.43	**82.28±0.47**

TABLE IX
MODEL COMPLEXITY AND OPERATIONAL EFFICIENCY OF METHODS ON HOUSTON 2013 DATASET

Indexes	FusAtNet	MDL	EndNet	S^2ENet	ExViT	HCTnet	Fus-MAE	NNCNet	SS-MAE	SCIIENet
Training time (s)	120.39	20.74	28.32	35.68	50.71	38.12	44.21	32.68	38.76	47.87
Inference time (s)	12.51	2.06	1.02	2.81	7.59	4.57	4.76	3.71	6.24	7.63
Parameters (M)	36.91	0.13	0.09	0.27	0.23	0.45	3.92	2.75	4.49	11.09
Model memory(MB)	605.05	2.11	18.57	21.46	21.02	24.56	142.94	61.71	89.74	209.97
FLOPs (M)	3462.62	2.4	0.46	13.26	45.84	35.93	331.62	76.12	429.93	654.96

the four datasets, the performance improvement is least noticeable in the Houston 2013 dataset, which contains the fewest samples. The potential of self-supervised methods has not been fully realized due to the limitation of unlabeled samples. In contrast, for the Berlin and Augsburg datasets, the performance of self-supervised methods is significantly better than that of supervised methods.

For self-supervised methods, SS-MAE preserves all features, including spatial and spectral information, by reconstruction but offers limited interaction. In contrast, NNCNet promotes interaction through CL but only fuses complementary information during the fine-tuning stage. As a result, NNCNet achieves performance comparable to that of SS-MAE. Although FusNet fuses features of the multimodal data for interaction during reconstruction, its performance is poorer than that of SS-MAE. This is primarily due to its inadequate consideration of spectral information and the limitations imposed by the significant gaps within the multimodal data during early fusion. In contrast, our proposed method employs CL combined with reconstruction to fully account for the shared and complementary information within multimodal data. This enhancement significantly improves the feature representation capability and classification performance of the model. In our proposed IES, we align the features, furthermore, using HCL before fusion to mitigate the effects of large differences between feature spaces. Additionally, we introduce the designed TCFE and MSCs to further reinforce interaction and reconstruction during the pretraining stage. Consequently, our method achieves the highest classification performance across all four datasets, demonstrating an increase in classification accuracy of 1.41% on Houston 2013, 4.12% on Augsburg, 4.53% on Berlin, and 2.07% on MUUFL relative to state-of-the-art methods.

2) Visual Evaluation: Figs. 7–10 sequentially present the visual classification results of various methods applied to the Houston 2013, Augsburg, Berlin, and MUUFL datasets, with distinct colors representing each category. The visualization results of our method align more closely with the ground-truth maps compared to other approaches, facilitating better differentiation of land cover classes and producing smoother classification maps. Specifically, for the Houston 2013 dataset, SCIIENet effectively delineates the boundaries between residential and commercial areas. In the Berlin dataset, it successfully distinguishes between commercial and residential areas. For the Augsburg dataset, our method achieves more accurate residential and industrial areas with fewer noise points. In the MUUFL dataset, it enhances the distinction between sidewalks and building roofs. Overall, the visualizations illustrate that our method effectively integrates HSIs with LiDAR/SAR data, leading to improved classification performance, particularly for classes with similar textural and spectral characteristics.

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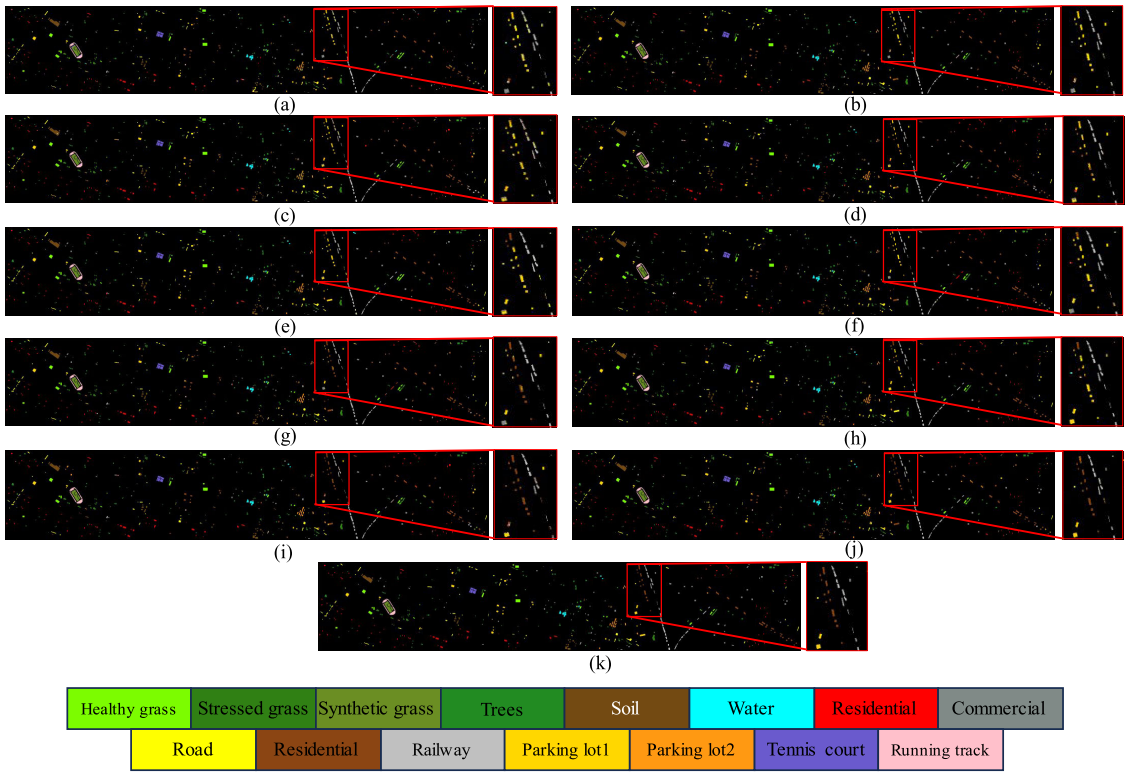


Fig. 7. Classification plots for different methods on the Houston 2013 dataset. (a) FusAtNet. (b) MDL. (c) EndNet. (d) S^{2}ENet. (e) ExViT. (f) HCTNet. (g) Fus-MAE. (h) NNCNet. (i) SS-MAE. (j) Proposed. (k) Ground-truth map.

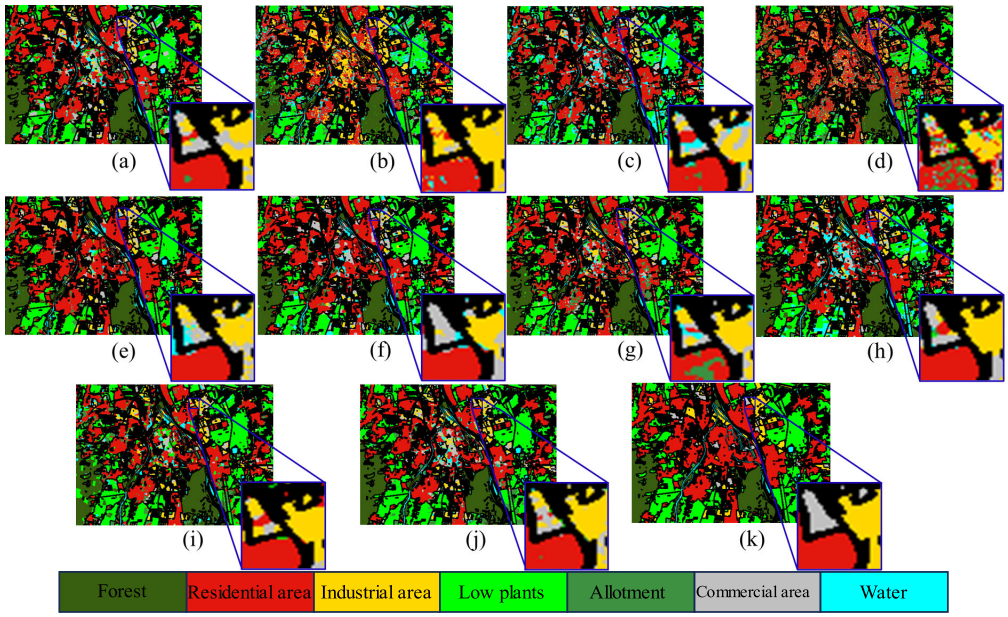


Fig. 8. Classification plots for different methods on the Augsburg dataset. (a) FusAtNet. (b) MDL. (c) EndNet. (d) S^{2}ENet. (e) ExViT. (f) HCTNet. (g) Fus-MAE. (h) NNCNet. (i) SS-MAE. (j) Proposed. (k) Ground-truth map.

3) Analysis of Computational Complexity: Table IX presents a comprehensive comparison of various models in terms of training time, inference time, number of parameters, floating point operations (FLOPs), and model memory on the Houston 2013 dataset. Specifically, the training time does not include the pretraining time. For these models, more complex models generally achieve superior classification accuracies, especially when dealing with challenging datasets. Supervised learning models typically have fewer parameters, model memory, and FLOPs than self-supervised models, as they do not require a pretraining stage. Among self-supervised methods, NNCNet exhibits fewer parameters, a smaller model memory, and lower FLOPs than MAE-based methods, primarily because it omits the reconstruction process that might lead to complementary information loss during pretraining. In contrast, Fus-MAE employs a single reconstruction branch, which limits its ability to effectively capture spectral information. SS-MAE manages one branch for spectral and one for spatial information. In order to effectively interact with the shared and complementary information of MRSIs, our proposed

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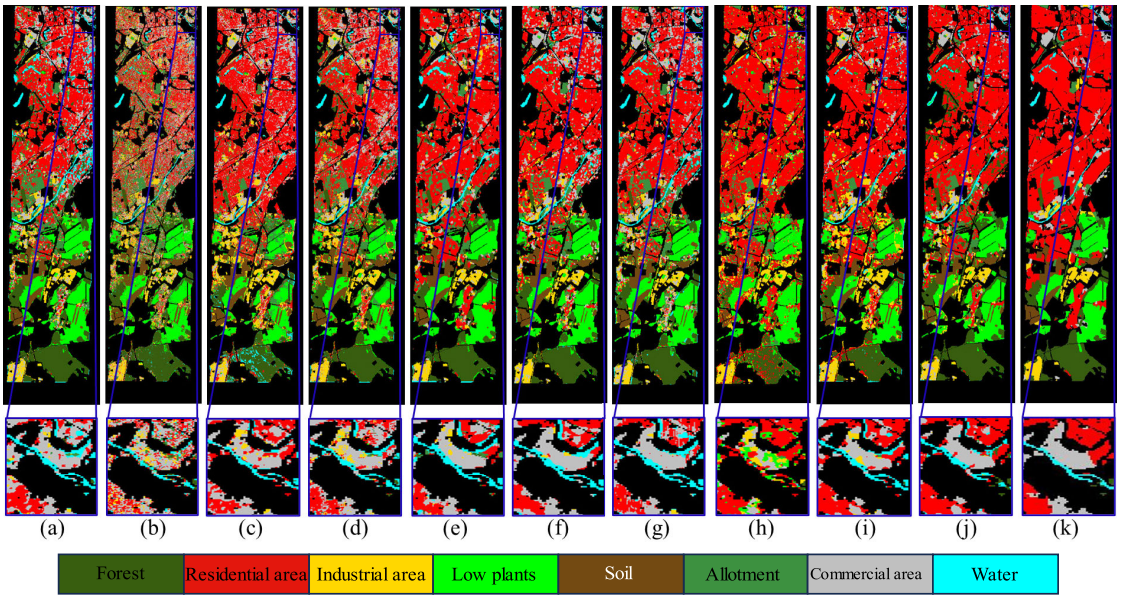


Fig. 9. Classification plots for different methods on the Berlin dataset. (a) FusAtNet. (b) MDL. (c) EndNet. (d) S^{2}ENet. (e) ExViT. (f) HCTNet. (g) Fus-MAE. (h) NNCNet. (i) SS-MAE. (j) Proposed. (k) Ground-truth map.

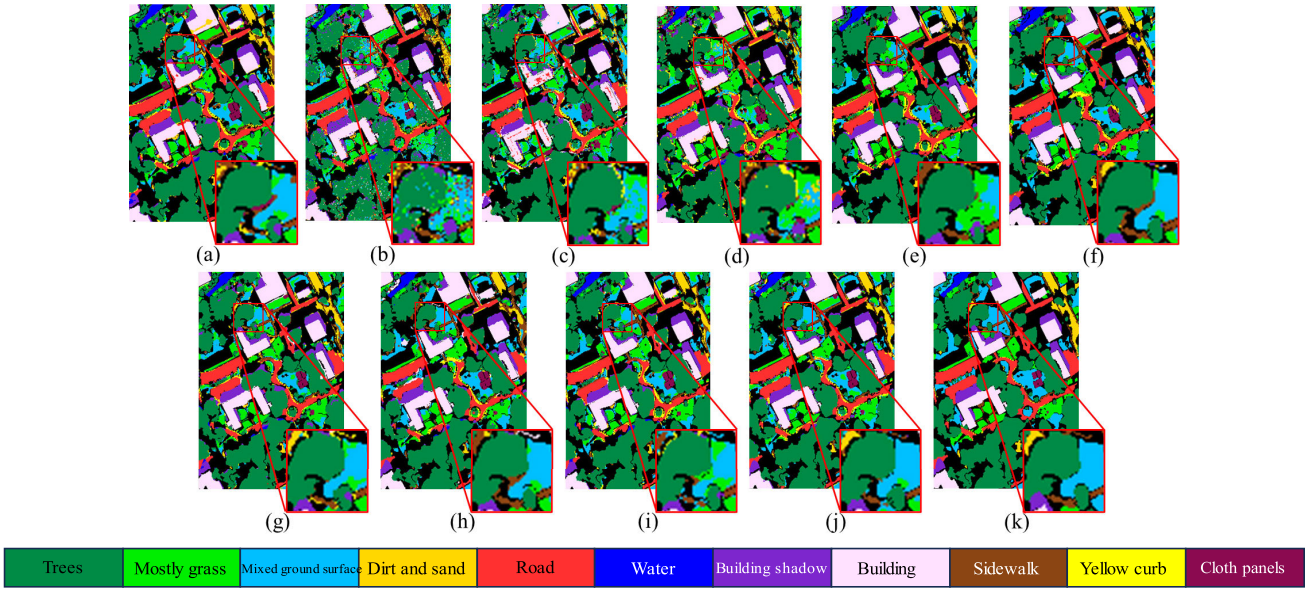


Fig. 10. Classification plots for different methods on the MUUFL dataset. (a) FusAtNet. (b) MDL. (c) EndNet. (d) S^{2}ENet. (e) ExViT. (f) HCTNet. (g) Fus-MAE. (h) NNCNet. (i) SS-MAE. (j) Proposed. (k) Ground-truth map.

SCIIENet integrates three reconstruction branches dedicated to addressing both spectral and spatial information. This architectural design results in SCIIENet having a relatively higher number of parameters and computational complexity compared to most models. It is, however, important to note that the reconstruction components do not participate in the actual training or inference processes. Consequently, the increased parameter count of SCIIENet does not significantly impact its training and inference time. Specifically, due to the pretraining stage, the network requires fewer training epochs compared to supervised models. For instance, although ExVit has fewer parameters, its lower learning efficiency necessitates more training epochs to achieve optimal performance. As a result, SCIIENet demonstrates significantly superior classification performance compared to other methods while maintaining comparable training and inference times with both ExVit and SS-MAE. This balance between performance and efficiency underscores the effectiveness of our proposed architecture.

E. Ablation Study

1) Ablation Analysis of the Number of Training Samples: Self-supervised learning effectively extracts features and has a reduced reliance on labeled samples. In order to evaluate the robustness of our model concerning sample size, we compare the classification performance of the fine-tuned model with and without pretraining across varying numbers of training samples, as illustrated in Fig. 11.

The base number is described in Section VI-A, along with 2-, 4-, 6-, 8-, and 10-fold variations to illustrate the performance differences. The results indicate that the proposed model significantly outperforms the model without pretraining when the number of training samples is small.

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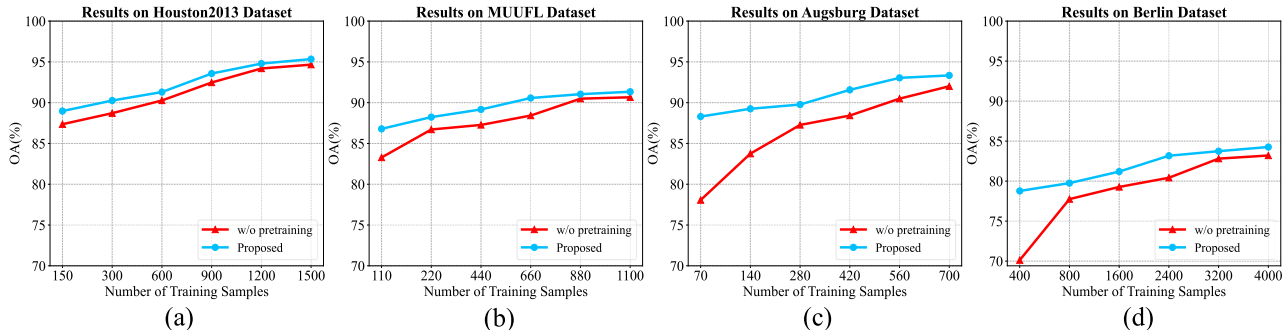


Fig. 11. Classification accuracy for different sample sizes. Results on (a) Houston 2013 dataset, (b) MUUFL dataset, (c) Augsburg dataset, and (d) Berlin dataset.

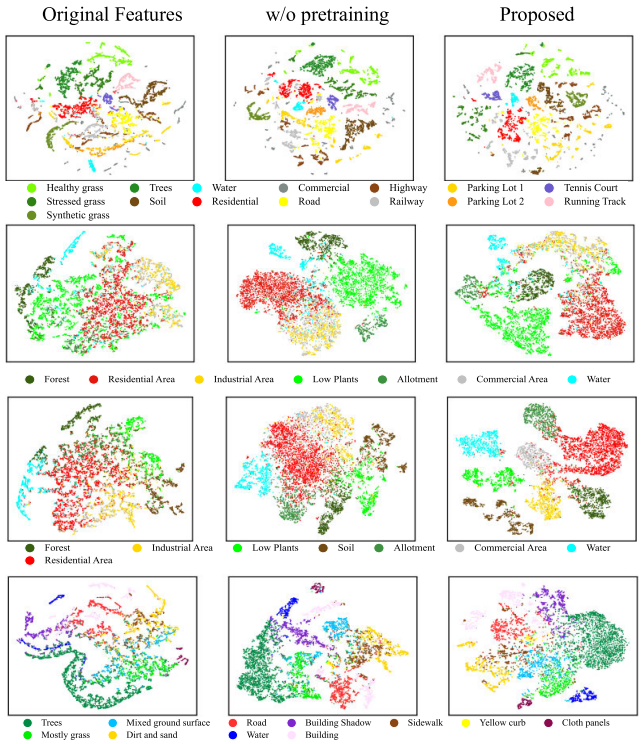


Fig. 12. Feature visualization of test samples from (top to bottom) Houston 2013, Augsburg, Berlin, and MUUFL datasets by t-SNE.

TABLE X

ABLATION STUDY ON TYPES OF CROSS-MODAL FUSION ENCODER

Type of fusion encoder	Metrics	Datasets			
		Houston 2013	Augsburg	Berlin	MUUFL
one-stage	OA(%)	89.56	87.37	77.49	86.13
	AA(%)	92.22	75.47	75.88	82.43
	$\kappa\times100$	89.04	82.27	67.96	81.91
two-stage	OA(%)	**90.18**	**87.57**	**78.82**	**86.61**
	AA(%)	**92.27**	**76.21**	**77.76**	**83.88**
	$\kappa\times100$	**89.38**	**82.77**	**68.25**	**82.28**

TABLE XI

ABLATION STUDIES OF LAYERS IN HCL

Number of layers in HCL	Metrics	Datasets			
		Houston 2013	Augsburg	Berlin	MUUFL
HCL^3(CL)	OA(%)	89.55	87.29	77.65	86.16
	AA(%)	91.74	75.95	76.09	82.56
	$\kappa\times100$	88.56	82.33	67.13	81.21
$HCL^{2,3}$	OA(%)	90.04	87.52	78.49	86.28
	AA(%)	91.97	76.17	77.74	82.89
	$\kappa\times100$	88.85	82.63	68.13	81.77
$HCL^{1,2,3}$	OA(%)	**90.18**	**87.57**	**78.82**	**86.61**
	AA(%)	**92.27**	**76.21**	**77.76**	**83.88**
	$\kappa\times100$	**89.38**	**82.77**	**68.25**	**82.28**

The performance gap between the two methods increases as the number of training samples decreases, particularly when the sample size falls below four times the base number. The supervised method requires 2–4 times more data to achieve performance comparable to that of the self-supervised approach. Notably, even with a ten-fold increase in sample size, the self-supervised method still outperforms the supervised one. We, furthermore, observe that the performance improvements gained through pretraining on the Augsburg and Berlin datasets are greater than those observed on the Houston 2013 and MUUFL datasets. This discrepancy can be attributed to the larger availability of unlabeled data in the Augsburg and Berlin datasets, facilitating more effective pretraining. Collectively, these results demonstrate that SCIIENet effectively alleviates the need for labeled data and optimally uses unlabeled data.

2) Feature Visualization: In order to further illustrate the effectiveness of self-supervised pretraining, the original data features alongside the features obtained after applying the fine-tuned model with and without pretraining using t-distributed stochastic neighbor embedding (t-SNE) is shown in Fig. 12.

We observe that after pretraining, the features of the same class are clustered closely together, while features from different classes are positioned farther apart. This suggests that the self-supervised model may be less likely to overfit than the supervised model. Additionally, we note that industrial, commercial, and residential areas are clustered in the second column but are distinctly separated in the third column. The substantial separability of these challenging categories in the third column further demonstrates the outstanding performance of our proposed method.

3) Stage of Cross-Modal Fusion Encoder: We conducted a comparative analysis between the TCFE and the one-stage structure, which solely employs MHCA, as depicted in configurations (8) and (9). The results of the ablation study, presented in Table X, highlight that the two-stage structure

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TABLE XII
ABLATION ANALYSIS OF DIFFERENT COMPONENTS IN SCIIENET

Model configuration						Metrics	Datasets			
CL	HCL	R	Spe	MSCs	TCFE		Houston 2013	Augsburg	Berlin	MUUFL
						OA(%)	87.37	81.88	74.63	83.17
✓						AA(%)	89.79	74.08	76.04	80.55
						$\kappa\times100$	86.36	75.66	64.49	78.47
						OA(%)	88.05	79.07	71.73	82.31
		✓				AA(%)	90.16	71.42	75.61	77.61
						$\kappa\times100$	87.09	72.21	60.28	76.55
						OA(%)	88.16	83.39	75.20	84.04
✓		✓				AA(%)	90.39	70.79	77.23	81.45
						$\kappa\times100$	87.19	77.32	64.73	79.04
						OA(%)	88.42	86.37	75.23	84.83
✓		✓	✓			AA(%)	90.52	75.33	76.76	81.81
						$\kappa\times100$	87.47	81.29	66.18	79.31
						OA(%)	89.02	86.99	77.48	85.74
✓		✓	✓	✓		AA(%)	91.05	74.76	75.67	82.41
						$\kappa\times100$	88.14	82.02	67.40	80.23
						OA(%)	89.55	87.29	77.65	86.16
✓		✓	✓	✓	✓	AA(%)	91.74	75.95	76.09	82.56
						$\kappa\times100$	88.56	82.33	67.13	81.21
						OA(%)	**90.18**	**87.57**	**78.82**	**86.61**
	✓	✓	✓	✓	✓	AA(%)	**92.27**	**76.21**	**77.76**	**83.88**
						$\kappa\times100$	**89.38**	**82.77**	**68.25**	**82.28**

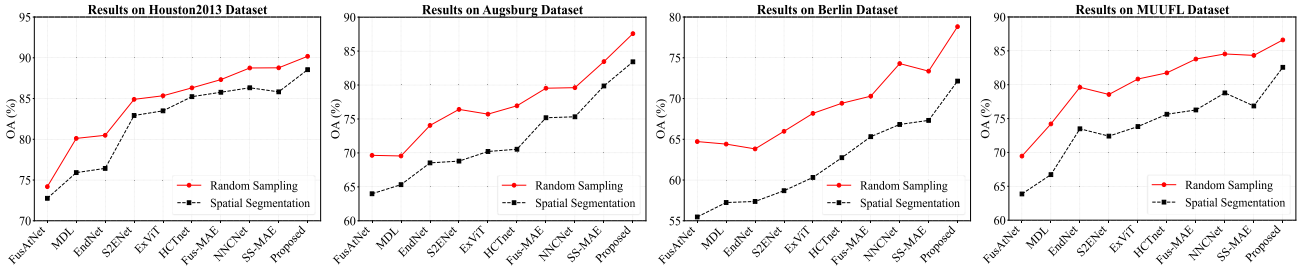


Fig. 13. Classification accuracy of each method under different sampling strategies. Results on (a) Houston 2013 dataset, (b) Augsburg dataset, (c) Berlin dataset, and (d) MUUFL dataset.

consistently outperforms the one-stage structure across all four datasets. This finding indicates that employing fused features, rather than features from individual modalities, contributes to more effective interaction enhancements. The fused features are less influenced by inaccuracies in captured data and contain more comprehensive and richer information compared to those derived from any single modality.

4) Effectiveness of HCL: In order to validate the effectiveness of HCL, we conducted ablation studies by applying CL to different layers, and the results are presented in Table XI. As indicated in the table, $HCL^{1,2,3}$ achieves the highest classification performance by applying CL across all three layers, whereas HCL^{3}, which applies CL only to the third layer, demonstrates the poorest performance. These findings demonstrate that HCL successfully mitigates the modality disparity by integrating features from all encoder layers, thus fostering a more effective feature fusion process in subsequent stages.

5) Effectiveness of Each Component: The components subjected to ablation in the proposed method include CL, reconstruction (R), the Spe branch, MSCs, HCL, and TCFE. In the complete ablation experiment, the CL and R as baseline networks are used as the baseline. The CL network incorporates contrastive loss applied to the final layer of the encoders in the Spa and L branches. On the other hand, the R network employs MAE for training the Spa and L branches. By sequentially embedding each network module, we conducted ablation analysis experiments across four datasets.

The specific results are presented in Table XII. When compared to the CL and R networks individually, the CL + R network exhibited significantly better classification performance across all four datasets. This indicates that the combination of CL and R is more effective in preserving complementary information while also aligning spatial features. Additionally, incorporating reconstruction during the pretraining stage is more effective for exploiting complementary information than solely fusing these features during the fine-tuning stage. While the CL + R network does not fully use spectral information, the CL + R + Spe network achieves enhanced performance by integrating the Spe branch, particularly evident in the Augsburg dataset. As we can find, the CL + R + Spe network outperforms existing self-supervised methods mentioned above, indicating that the triple-branch framework, which harmonizes CL and R to comprehensively account for shared and complementary spatial information, and complementary spectral information, is effective. The CL + R + Spe + MSCs network, furthermore, leverages multiple intermediate features to enhance the reconstruction process,

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further preserving complementary information and improving classification accuracy. Finally, by integrating the IES components, the SCIIENet is obtained. By enhancing interaction through HCL and TCFE, the shared and complementary information from different modalities is effectively integrated, thereby fulfilling the requirements for precise classification. In conclusion, the SCIIENet achieves state-of-the-art performance through this combination of modules, providing innovative strategies for current research in the classification of multimodal data.

F. Influence of Spatial Correlation on Classification Results

Although random sampling is commonly employed in the literature to evaluate classification performance, spatial correlation inherent in random sampling can lead to information leakage between the training and test sets, and the performance of methods is overestimated. In order to mitigate this issue and validate the robustness of our proposed method, we adopted a specific spatial sampling strategy designed to minimize the influence of spatial correlation. Specifically, the image was divided into 10×10 nonoverlapping blocks, and random sampling was confined to approximately 20% of these blocks for each category. This approach effectively reduces spatial correlation between training and test samples compared to conventional random sampling.

The classification performance of various methods under this spatial sampling strategy is illustrated in Fig. 13. Despite the application of very low sampling rates in random sampling, spatial correlation still results in an overestimation of the performance of these methods. In contrast, SCIIENet continues to outperform other methods significantly under spatial segmentation, with the performance gap widening further. SCIIENet maintains superior performance with improvements of 2.22% (Houston 2013: 88.54% versus 86.32%), 3.56% (Augsburg: 83.42% versus 79.86%), 4.82% (Berlin: 72.14% versus 67.32%), and 3.78% (MUUFL: 82.57% versus 78.79%) over suboptimal methods. These results demonstrate SCIIENet's robust feature extraction capability, attributed to the designed pretraining architecture. Its consistent performance ensures that the method's effectiveness is not reliant on spatial correlation, making it highly suitable for real-world applications in large-scale scenes.

V. Conclusion

In this study, a SCIIENet is proposed for MRSI classification. This method fully considers the complementary spectral information, shared spatial information, and complementary spatial information through a triple-branch reconstruction structure with CL. Additionally, we introduce an IES that uses HCL to better align multimodal features and incorporates a TCFE to enhance both shared and complementary information for interaction. We employ MSCs, furthermore, to preserve complementary features. The proposed method effectively improves the performance of MRSI classification, as demonstrated by extensive experiments on the Houston 2013, Augsburg, Berlin, and MUUFL datasets, which highlight the superiority of our model over state-of-the-art methods.

While our study primarily focuses on cross-modal information, which is a critical feature of multimodal data, exploring intramodal information could further enhance classification performance. For example, in future work, we can incorporate CL for each modality in the proposed model. Extending our method to incorporate more than two modalities could, furthermore, potentially enhance classification performance and applicability.

References

[1] Z. Li, L. Huang, and J. He, "A multiscale deep middle-level feature fusion network for hyperspectral classification," *Remote Sens.*, vol. 11, no. 6, p. 695, Mar. 2019.

[2] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "CoSpace: Common subspace learning from hyperspectral-multispectral correspondences," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4349–4359, Jul. 2019.

[3] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[4] M. Khodadadzadeh, J. Li, S. Prasad, and A. Plaza, "Fusion of hyperspectral and LiDAR remote sensing data using multiple feature learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2971–2983, Jun. 2015.

[5] B. Rasti, P. Ghamisi, J. Plaza, and A. Plaza, "Fusion of hyperspectral and LiDAR data using sparse and low-rank component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6354–6365, Nov. 2017.

[6] W. Yu, H. Huang, M. Zhang, Y. Shen, and G. Shen, "Shadow mask-driven multimodal intrinsic image decomposition for hyperspectral and LiDAR data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5525915.

[7] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, Mar. 2013.

[8] B. Huang, Y. Li, X. Han, Y. Cui, W. Li, and R. Li, "Cloud removal from optical satellite imagery with SAR imagery using sparse representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 5, pp. 1046–1050, May 2015.

[9] L. Gómez-Chova, D. Tuia, G. Moser, and G. Camps-Valls, "Multimodal classification of remote sensing images: A review and future directions," *Proc. IEEE*, vol. 103, no. 9, pp. 1560–1584, Sep. 2015.

[10] F. Jahan, J. Zhou, M. Awrangjeb, and Y. Gao, "Integration of heterogeneous features from co-registered hyperspectral and LiDAR data for land cover classification," in *Proc. 10th IAPR Workshop Pattern Recognit. Remote Sens. (PRRS)*, Beijing, China, Aug. 2018, pp. 1–6.

[11] A. Ibrahim et al., "Atmospheric correction for hyperspectral ocean color retrieval with application to the hyperspectral imager for the coastal ocean (HICO)," *Remote Sens. Environ.*, vol. 204, pp. 60–75, Jan. 2018.

[12] A. Kuras, M. Brell, S. Teien, I. Burud, and T. Thiis, "Feature-level based hyperspectral and LiDAR data fusion for urban analysis," in *Proc. Joint Urban Remote Sens. Event (JURSE)*, Heraklion, Greece, May 2023, pp. 1–4.

[13] J. Li, K. Zheng, W. Liu, Z. Li, H. Yu, and L. Ni, "Model-guided coarse-to-fine fusion network for unsupervised hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[14] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.

[15] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[16] J. Li, K. Zheng, J. Yao, L. Gao, and D. Hong, "Deep unsupervised blind hyperspectral and multispectral data fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[17] Y. Chen, C. Li, P. Ghamisi, X. Jia, and Y. Gu, "Deep fusion of remote sensing data for accurate classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1253–1257, Aug. 2017.

[18] Y. Zhang et al., "A cross-modal feature aggregation and enhancement network for hyperspectral and LiDAR joint classification," *Expert Syst. Appl.*, vol. 258, Dec. 2024, Art. no. 125145.

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[19] J. Wang, J. Li, Y. Shi, J. Lai, and X. Tan, "AM^{3}net: Adaptive mutual-learning-based multimodal data fusion network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 8, pp. 5411–5426, Aug. 2022.

[20] W. He, X. Fu, N. Li, Q. Ren, and S. Jia, "LGCT: Local–global collaborative transformer for fusion of hyperspectral and multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5537114.

[21] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[22] H. Wang, Y. Xu, Z. Wu, and Z. Wei, "Unsupervised hyperspectral and multispectral image blind fusion based on deep tucker decomposition network with spatial–spectral manifold learning," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Oct. 4, 2024, doi: 10.1109/TNNLS.2024.3457781.

[23] S. Falahatnejad and A. Karami, "Deep fusion of hyperspectral and LiDAR images using attention-based CNN," *Social Netw. Comput. Sci.*, vol. 4, no. 1, Oct. 2022.

[24] S. Chen, L. Zhang, and L. Zhang, "Cyclic cross-modality interaction for hyperspectral and multispectral image fusion," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 1, pp. 741–753, Jan. 2025.

[25] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010.

[26] X. Du, X. Zheng, X. Lu, and A. A. Doudkin, "Multisource remote sensing data classification with graph fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10062–10072, Dec. 2021.

[27] W. Zhou and Seiichiro-Kamata, "Multi-scanning based recurrent neural network for hyperspectral image classification," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Milan, Italy, Jan. 2021, pp. 4743–4750.

[28] X. Song et al., "MGPACNet: A multiscale geometric prior aware cross-modal network for images fusion classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4412815.

[29] L. Jing and Y. Tian, "Self-supervised visual feature learning with deep neural networks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 11, pp. 4037–4058, Nov. 2021.

[30] B. Ren, Y. Zhao, B. Hou, J. Chanussot, and L. Jiao, "A mutual information-based self-supervised learning model for PolSAR land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9224–9237, Nov. 2021.

[31] R. Hang, X. Qian, and Q. Liu, "Cross-modality contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532812.

[32] H. Wu, S. Dai, C. Liu, A. Wang, and Y. Iwahori, "A novel dual-encoder model for hyperspectral and LiDAR joint classification via contrastive learning," *Remote Sens.*, vol. 15, no. 4, p. 924, Feb. 2023.

[33] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816.

[34] Z. Feng, L. Song, S. Yang, X. Zhang, and L. Jiao, "Cross-modal contrastive learning for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5517713.

[35] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, USA, Jun. 2022, pp. 15979–15988.

[36] Z. Xue, X. Yu, A. Yu, B. Liu, P. Zhang, and S. Wu, "Self-supervised feature learning for multimodal remote sensing image land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533815.

[37] J. Lin, F. Gao, X. Shi, J. Dong, and Q. Du, "SS-MAE: Spatial–spectral masked autoencoder for multisource remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531614.

[38] H. Chan-To-Hing and B. Veeravalli, "FUS-MAE: A cross-attention-based data fusion approach for masked autoencoders in remote sensing," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Athens, Greece, Jul. 2024, pp. 6953–6958.

[39] H. Gao, H. Feng, Y. Zhang, S. Xu, and B. Zhang, "AMSSE-net: Adaptive multiscale spatial–spectral enhancement network for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531317.

[40] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.

[41] T. Song, Z. Zeng, C. Gao, H. Chen, and J. Li, "Joint classification of hyperspectral and LiDAR data using height information guided hierarchical fusion-and-separation network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505315.

[42] X. Wang, Y. Feng, R. Song, Z. Mu, and C. Song, "Multi-attentive hierarchical dense fusion net for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 82, pp. 1–18, Jun. 2022.

[43] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[44] S. Hou, H. Shi, X. Cao, X. Zhang, and L. Jiao, "Hyperspectral imagery classification based on contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521213.

[45] D. L. Ayuba, J.-Y. Guillemaut, B. Marti-Cardona, and O. Mendez, "HyperKon: A self-supervised contrastive network for hyperspectral image analysis," *Remote Sens.*, vol. 16, no. 18, p. 3399, Sep. 2024.

[46] Y. Qin et al., "Nearest neighboring self-supervised learning for hyperspectral image classification," *Remote Sens.*, vol. 15, no. 6, p. 1713, Mar. 2023.

[47] S. Xia, X. Zhang, H. Meng, J. Fan, and L. Jiao, "Two-stream networks for contrastive learning in hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 1903–1920, 2024.

[48] W. Liu et al., "Self-supervised feature learning based on spectral masking for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4407715.

[49] D. Ibañez, R. Fernandez-Beltran, F. Pla, and N. Yokoya, "Masked auto-encoding spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5542614.

[50] S. Mohamed, M. Haghighat, T. Fernando, S. Sridharan, C. Fookes, and P. Moghadam, "FactoFormer: Factorized hyperspectral transformers with self-supervised pretraining," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5501614.

[51] L. Huang, Y. Chen, and X. He, "Spectral–spatial masked transformer with supervised and contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508718.

[52] X. Cao, H. Lin, S. Guo, T. Xiong, and L. Jiao, "Transformer-based masked autoencoder with contrastive loss for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524312.

[53] Z. Xue, B. Liu, A. Yu, X. Yu, P. Zhang, and X. Tan, "Self-supervised feature representation and few-shot land cover classification of multimodal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541618.

[54] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2018, *arXiv:1807.03748*.

[55] A. Vaswani, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 6000–6010.

[56] J. X. Yang, J. Zhou, J. Wang, H. Tian, and A. W.-C. Liew, "LiDAR-guided cross-attention fusion for hyperspectral band selection and image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5515815.

[57] Y. Hu, H. He, and L. Weng, "Hyperspectral and LiDAR data land-use classification using parallel transformers," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 703–706.

[58] Y. Liu, S. Zhang, J. Chen, Z. Yu, K. Chen, and D. Lin, "Improving pixel-based MIM by reducing wasted modeling capability," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Paris, France, Oct. 2023, pp. 5338–5349.

[59] S. Mohla, S. Pande, B. Banerjee, and S. Chaudhuri, "FusAtNet: Dual attention based SpectroSpatial multimodal fusion network for hyperspectral and LiDAR classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Seattle, WA, USA, Jun. 2020, pp. 416–425.

[60] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[61] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[62] S. Fang, K. Li, and Z. Li, "S^{2}ENet: Spatial–spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[63] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.

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